

The Variance Risk Premium as a Return Predictor

Replication of Bollerslev, Tauchen & Zhou (2009)

Expected Stock Returns and Variance Risk Premia, Review of Financial Studies 22(11), 2009

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- $VRP = IV - RV = VIX^2/12 - \Sigma r^2_d$ (variance risk premium construction)
 - Epstein–Zin recursive preferences: why volatility-of-volatility creates return predictability
 - Horizon regressions $h = 1\text{--}24$ months with Newey–West standard errors
 - Sign reversal in 2008–2026: the paper’s result does not survive out-of-sample

Course Roadmap

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The VRP definition and its equilibrium foundations (Epstein–Zin)

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Theory

The VRP and Its Equilibrium Foundations

What Is the Variance Risk Premium?

The VRP: Plain-Language Definition

- The VIX is the market's price of variance insurance
- Variance swap: pay IV today, receive RV that materialises
- If IV > RV consistently, investors pay a fear premium
- $VRP_t = IV_t - RV_t$ (positive = market over-prices variance)

Key Formula

$$VRP_t = IV_t - RV_t$$

$$IV_t = VIX_t^2 / 12 \quad (\%^2 / \text{month})$$

$$RV_t = \Sigma^d [100 \times \ln(P^d / P_{-1}^d)]^2$$

Units: %² per month (variance, not volatility)

8.99 %²

Mean VRP (2008–2026)

85.2%

Months with VRP > 0

−0.10

VRP AR(1) [paper: 0.49]

47.3 %²

VRP Std Dev [paper: 15.1]

Our Sample vs. Paper

- Paper (1990–2007): mean VRP = 18.30, AR(1) = 0.49 — persistent signal
- Ours (2008–2026): mean VRP = 8.99, AR(1) = −0.10 — near-random walk
- Two crises (2008, 2020) dominate the distribution

Why Does VRP Predict Returns? Epstein–Zin Theory

Equity Premium Decomposition (Bollerslev et al. Eq. 11)

$$\pi_{r,t} = \gamma \sigma_{\{g,t\}}^2 + (1 - \theta) \kappa^1 (A_q^2 \varphi_q^2 + A_\sigma^2) \times q_t$$

[consumption risk] + [volatility-of-volatility risk]

q_t = volatility of the variance process | γ = risk aversion | θ = EIS parameter

The VRP Isolates q_t

- $\text{VRP}_t = E^T(\text{RV}_{\{t+1\}}) - E(\text{RV}_{\{t+1\}})$ measures the risk-neutral vs. physical variance gap
- This gap is exactly proportional to q_t (vol-of-vol), the premium's second term

Why the Quarterly Horizon?

- At $h=1$: too noisy — monthly returns have low signal-to-noise
- At $h=3$: optimal — within q_t persistence window, noise averaged
- At $h=12$: q_t persistence fades, signal decays toward zero
- Paper peak: $h=3$, $t=2.86$, $R^2=6.82\%$

Model Assumptions (and Our Departures)

- Model requires: (1) EZ preferences with $\text{IES} > 1$ and risk aversion > 1 , (2) stochastic vol-of-vol process q_t , (3) persistent VRP ($\text{AR1} \approx 0.5$)
- Our data: VRP $\text{AR1} = -0.10$ — near-zero persistence violates assumption (3). The model's prediction is conditional on q_t being persistent.

Paper's Key Findings and What We Can Replicate

Horizon h	Slope b_1	t-stat (NW)	Adj. R^2	Interpretation
1 month	0.39	1.76	1.07%	Weakly positive
3 months	+0.47	+2.86	6.82%	PEAK — strong quarterly signal
6 months	0.30	2.15	5.42%	Significant but fading
12 months	0.12	1.00	1.23%	Weak, near zero

What We CAN Replicate

- VRP construction methodology (IV and RV formulas)
- Horizon regression framework (Equation 17)
- Qualitative structure of the test (NW standard errors)

What We CANNOT Replicate

- 1990–2007 sample: SPX data starts Oct 28, 2008
- 5-minute intraday RV: we have daily data only
- 216 months of paper's calm, persistent VRP regime

2

Signal Construction

Computing IV, RV, and the Variance Risk Premium

Data Sources and Sample Coverage

Dataset	Coverage	Role in Analysis	Key Limitation
spx_prices.parquet	Oct 28 2008 – Feb 2026	Monthly RV via daily log-returns	Starts mid-crisis; Oct 2008 = 4 days only
vix_historical.parquet	Jan 2000 – Mar 2026	Monthly IV = $VIX^2/12$	End-of-month sampling discards intra-month path
interest_rates.parquet	1954 – 2026	Risk-free rate proxy (Fed Funds / 12)	Policy rate $\neq$ 3-month T-bill (paper's rate)

209

months in sample

Oct 2008

start (vs. 1990 in paper)

4 days

Oct 2008 trading days

Feb 2026

end of sample

Critical Data Artifact — October 2008

- Our SPX series starts Oct 28, 2008. That month has only 4 trading days $\rightarrow$ $RV = 10.1\% ^2$ (artificially tiny).
- VIX closed at 59 that month $\rightarrow$ $IV = 59^2/12 = 290$. $VRP = +289\% ^2$ — the highest in our sample, but entirely a data artifact.

Realized Variance and Implied Variance: Construction

Realized Variance (Daily Approximation)

$$RV_t = \sum_{\{d \in \text{month } t\}} [100 \times \ln(P_d / P_{\{d-1\}})]^2$$

Units: %² per month

Paper: 5-min intraday, $n = 22 \times 78 = 1,716$ obs

Ours: daily closes, $n \approx 22$ obs per month

Our mean RV = 29.52 %² vs paper 14.93 %²

Implied Variance (from VIX)

$$IV_t = VIX_{\{t, EOM\}}^2 / 12$$

VIX = annualized implied vol in %

VIX = 20 $\Rightarrow$ IV = 400/12 = 33.3 %²/mo

Paper mean IV = 33.23 (1990–2007)

Our mean IV = 38.50 (2008–2026)

End-of-month sampling (last VIX close)

Why Daily RV is Noisier

- 22 observations vs 1,716 — sampling variance is $\approx 75\times$ higher
- October 2008 artifact (4 obs) would be impossible with 5-min data

Unit Reconciliation (Both in %²/month)

- $VIX^2/12 \Rightarrow$ annualized %² $\div 12 =$ monthly %² $\checkmark$
- Eq. (22): $VRP_t = VIX_{\{t\}}^2/12 - \sum_{\{d \in t\}} r_{\{d, \%\}}^2$
- Both terms in identical units — no scaling needed

No-Lookahead Check

- VRP_t uses only VIX and SPX prices within month t — fully known at month-end t , before $t+1$ begins.
- Regression: forward return at $t+1$ on VRP_t at t . Position computed at t applied to return at $t+1$. $\checkmark$

VRP Assembly and Summary Statistics vs. Paper

Key Formulas

$$VRP_t = IV_t - RV_t = VIX_t^2 / 12 - \Sigma_{\{d \in t\}} r_{\{d, \%\}}^2$$

$$R^e_{\{t+1\}} = \ln(SPX_{\{t+1\}} / SPX_t) - r_{\{f, t\}} \quad (\text{monthly excess return})$$

Variable	Our Mean	Paper Mean	Our Std	Paper Std	Our AR(1)	Paper AR(1)
Excess Ret. (ann. %)	10.08	6.44	51.50	47.19	-0.08	-0.03
VRP (% ² /mo)	8.99	18.30	47.33	15.13	-0.10	0.49
IV (% ² /mo)	38.50	33.23	40.44	23.73	0.78	0.79
RV (% ² /mo)	29.52	14.93	63.95	15.25	0.30	0.70

Key Differences Driving Failed Replication

- VRP AR(1) = -0.10 (ours) vs 0.49 (paper): the signal has no persistence. Each month is essentially independent.
- RV std = 63.95 (ours) vs 15.25 (paper): dominated by the March 2020 extreme (RV = 747 %²).

Implied Variance, Realized Variance, and VRP Over Time (2008–2026)

Figure 2 Analog — S&P 500 Implied and Realized Variance (2008–2026)

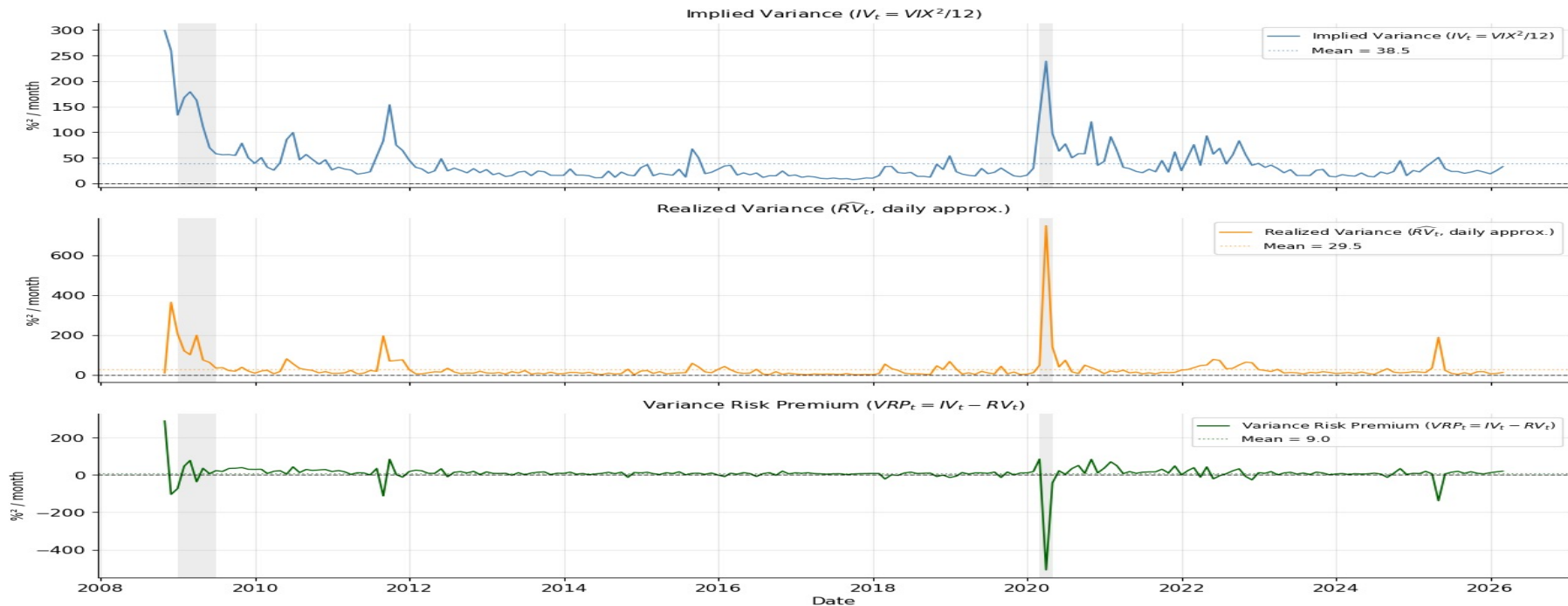


Fig. 2 analog: IV (top) = $VIX^2/12$; RV (mid) = daily approximation; VRP (bottom) = $IV - RV$. Grey shading = NBER recessions. Dashed line = sample mean.

3

Statistical Validation

Horizon Regressions and the Sign Reversal

The Horizon Regression: Equation (17)

Core Predictive Regression — Bollerslev et al. Equation (17)

$$\left(\frac{1}{h}\right) \sum_{j=1}^{\{h\}} r_{\{t+j\}}^e = b_0(h) + b_1(h) \cdot VRP_t + u_{\{t+h,t\}}$$

$b_1(h) > 0$ is the paper's prediction: high VRP today $\Rightarrow$ high returns over next h months

Forward Return Construction

- `returns.rolling(h).mean().shift(-h)` at index t gives the mean of $r_{\{t+1\}}, \dots, r_{\{t+h\}}$
- Fully out-of-sample relative to VRP_t which is known at t
- Overlapping windows for $h > 1$ — creates $MA(h-1)$ residual autocorrelation

Standard Errors (Newey–West HAC)

- Paper uses Hodrick (1992) reverse-regression SE
- We use Newey–West with `maxlags = h` (similar correction)
- Both correct for $MA(h-1)$ overlap bias in overlapping returns
- NW SE tend to be slightly more conservative than Hodrick

Interpreting the Results

- $b_1 \approx 0.47$ at $h=3$ (paper): a 1 %² increase in VRP $\rightarrow$ +0.47% higher quarterly return
- $R^2 = 6.82\%$ at $h=3$ (paper): VRP explains 6.82% of variation in 3-month forward returns
- The hump shape — R^2 rising then falling with h — is the paper's diagnostic signature

Table 2 Analog: VRP→Return Regression Results

Horizon h	Paper b_1	Our b_1	Paper t-stat	Our t-stat	Our R^2 (%)	Sign
h = 1	+0.39	-0.012	+1.76	-1.24	1.12%	OPPOSITE
h = 3	+0.47	-0.005	+2.86	-0.96	0.42%	OPPOSITE
h = 6	+0.30	-0.004	+2.15	-1.64	0.98%	OPPOSITE
h = 9	+0.17	-0.003	+1.36	-1.82	1.54%	OPPOSITE
h = 12	+0.12	-0.003	+1.00	-2.18	1.76%	OPPOSITE

The Slope Sign is Negative at Every Single Horizon

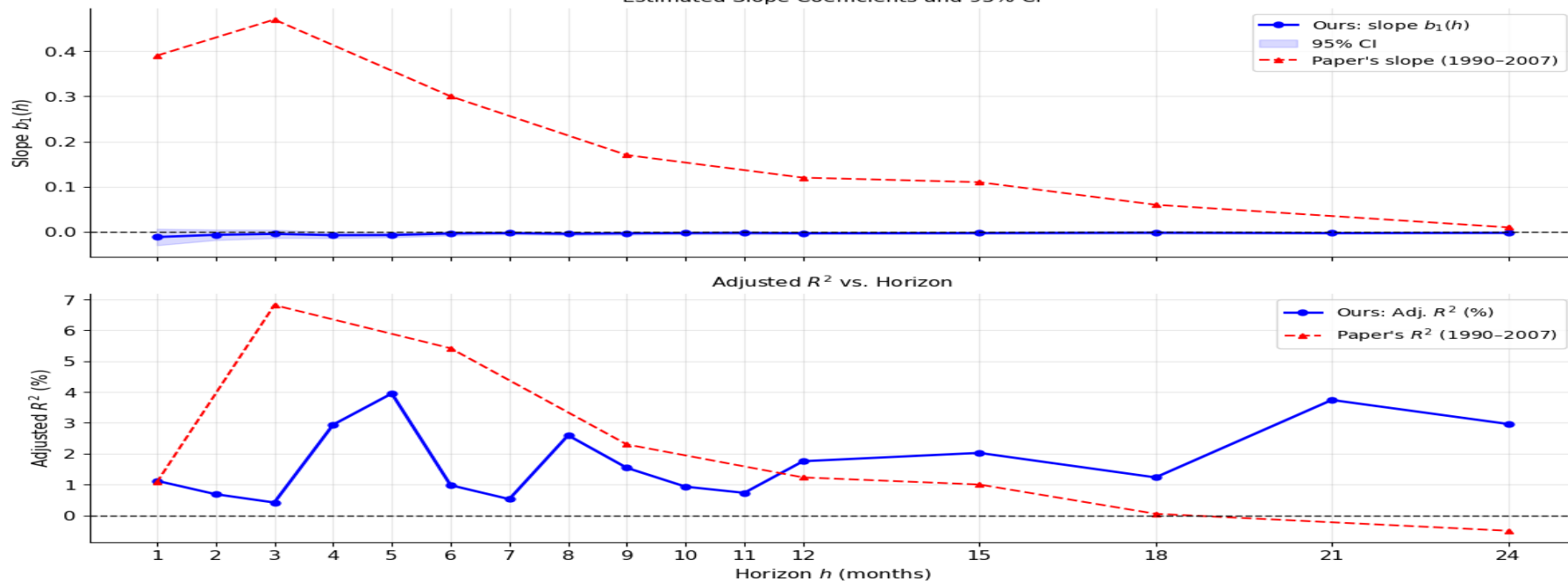
- Paper: b_1 positive and hump-shaped, peaking at $h=3$. Ours: b_1 negative and monotonically increasing toward zero.
- The direction of the result has completely reversed — this is not a magnitude difference but a qualitative failure to replicate.

Ours: 2008–10 to 2026–02, $N=209$ months, Newey–West SE ($\text{maxlags}=h$). Paper: 1990–01 to 2007–12, $N=216$ months, Hodrick (1992) SE.

Estimated Slopes and R^2 Across Horizons: Paper vs. Ours

Figure 3 Analog — Paper (red) vs. Our Replication (blue)

Estimated Slope Coefficients and 95% CI



Top: Slope $b_1(h)$ with 95% CI — paper (red $\blacktriangle$) is positive at all horizons; ours (blue $\bullet$) is negative. Bottom: Adj. R^2 — paper peaks at $h=3$ (6.82%); ours is flat.

What Do We Actually See? The Sign Reversal Explained

Paper: 1990–2007 Finding

- $h=3$: $b_1 = +0.47$, $t = +2.86$, $R^2 = 6.82\%$
- Positive at every horizon: high VRP $\Rightarrow$ high returns
- Hump-shaped R^2 : peaks quarterly, decays at 12m
- VRP $AR(1) = 0.49$: signal is persistent, predictable

Ours: 2008–2026 Finding

- $h=3$: $b_1 = -0.005$, $t = -0.96$, $R^2 = 0.42\%$
- Negative at every horizon: high VRP $\Rightarrow$ bad returns
- Flat R^2 profile (0.4%–3%), no quarterly hump
- VRP $AR(1) = -0.10$: near-zero persistence

Three Observation Pairs Drive the Sign Reversal

- Oct 2008 (VRP = +289, data artifact: 4 trading days only) $\rightarrow$ Nov 2008 = -7.9% : high VRP $\rightarrow$ big loss
- Feb 2020 (VRP = +85, COVID fear spiked VIX before the crash) $\rightarrow$ Mar 2020 = -13.4% : high VRP $\rightarrow$ disaster
- Mar 2020 (VRP = -509 , realized vol exploded past implied) $\rightarrow$ Apr 2020 = $+11.9\%$: low VRP $\rightarrow$ big gain

Key Takeaway: Regime Change in VRP Timing

- 1990–2007: fear spikes (high VRP) were forward-looking — followed by recovery. 2008–2026: fear spikes coincide with crashes already in progress, then recoveries follow the low-VRP crash month.

4

Position Sizing

Translating VRP into Trading Signals

Position Sizing: Binary and Scaled VRP Signals

Binary Signal

```
positiont = 1   if VRPt > 0
positiont = 0   otherwise
```

Logic: long S&P500 when 'fear' is positive

Scaled Signal (Z-score)

```
zt = (VRPt - μ{36}) / σ{36}
positiont = clip(zt, 0, 2) / 2
```

Logic: proportional to how extreme VRP is

Signal	VRP > 0 fraction	Avg Position	VRP>0 months: return/mo	VRP<0 months: return/mo
Binary	85.2% (178/209 months)	0.85	+1.03% (+12.4% ann.)	-0.27% (-3.2% ann.)
Scaled	Same VRP sign	0.10	(proportional)	(proportional)

No-Lookahead Enforcement

- position_t is computed at month-end t
- Applied to return r_{t+1} via .shift(1) in backtest
- VRP_t uses only data available at t ✓

Conditional Returns (Contemporaneous)

- VRP>0 months average +1.03%/mo — but this is same-month!
- Contemporaneous comparison ≠ predictive regression
- Next-month conditioning is the correct test (Section 3)

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Backtest

Portfolio Performance: VRP Timing vs. Buy & Hold

Backtest Performance: 2008–10 to 2026–02 (N = 209 months)

Metric	Buy & Hold	VRP Binary	VRP Scaled	Binary vs. B&H
Ann. Return (%)	10.08	9.60	0.87	−0.48%
Ann. Volatility (%)	14.87	12.84	4.67	−2.03%
Sharpe Ratio	0.68	0.75	0.19	+0.07
Max Drawdown (%)	−29.4	−22.6	−20.9	+6.8 pp
Hit Rate (%)	64.6	55.3	33.0	−9.3 pp

0.75

VRP Binary Sharpe

0.68

Buy & Hold Sharpe

−22.6%

VRP Binary Max DD

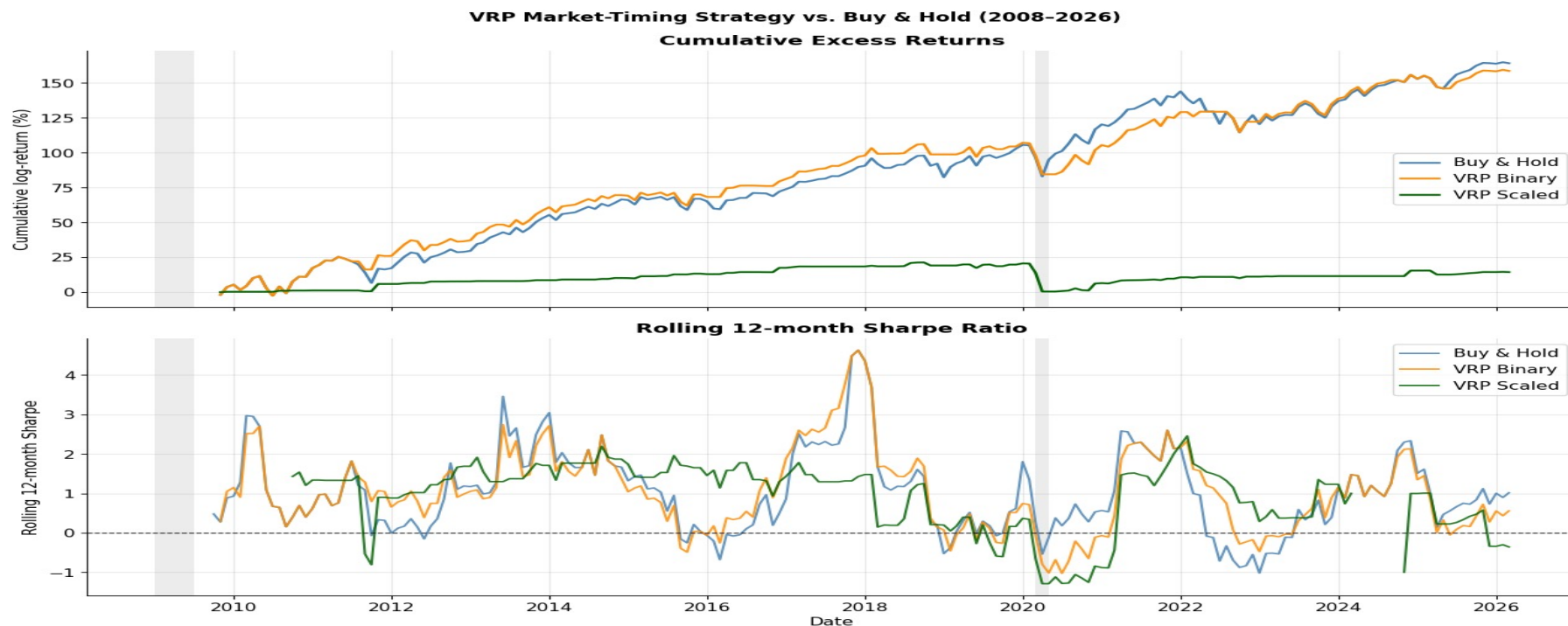
−29.4%

Buy & Hold Max DD

Mechanism: Tail-Risk Filter, Not Return Predictor

- Strategy is flat 14.8% of months (VRP<0). Those 31 months include Nov 2008, Mar 2020, Aug 2011 — the worst episodes. Avoiding them reduces the left tail. Hit rate drops (55.3% vs 64.6%) because up-months are also missed. Net: modest Sharpe lift.

Cumulative Returns and Rolling 12-Month Sharpe Ratio



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Paper Comparison

Table 2 Replication: Slopes, t-stats, and R^2

Table 2 Replication: Bollerslev et al. (1990–2007) vs. Ours (2008–2026)

h	Paper b_1	Our b_1	Paper t	Our t	Paper $R^2\%$	Our $R^2\%$
1	+0.39	-0.012	+1.76	-1.24	1.07	1.12
3	+0.47	-0.005	+2.86	-0.96	6.82	0.42
6	+0.30	-0.004	+2.15	-1.64	5.42	0.98
9	+0.17	-0.003	+1.36	-1.82	2.30	1.54
12	+0.12	-0.003	+1.00	-2.18	1.23	1.76

Green = positive (paper direction). Red = negative (our direction). $H=3$ is the paper's peak: R^2 drops from 6.82% to 0.42% in our sample.

Shape Comparison

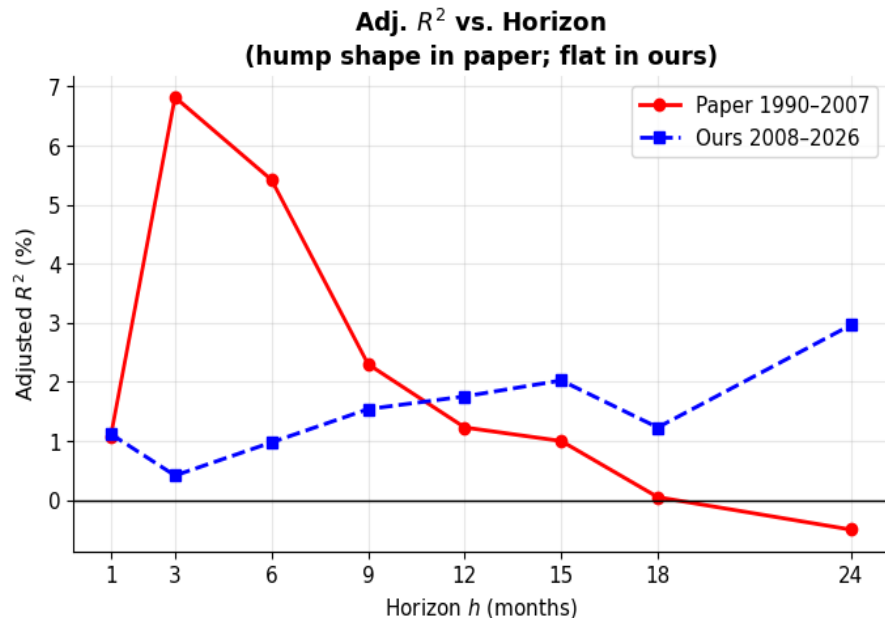
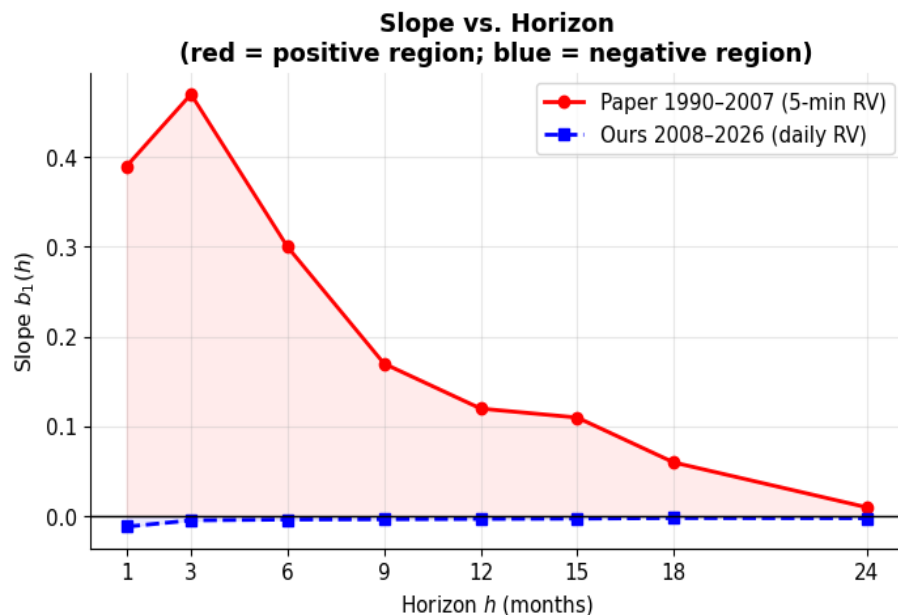
- Paper: hump-shaped R^2 peaking at $h=3$, then decaying
- Ours: flat/slightly rising R^2 from $h=1$ to $h=24$ (0.4%→3.0%)
- Both the magnitude AND the shape fail to replicate

Causes

- (1) Wrong sample epoch — dominant driver
- (2) Oct 2008 partial-month artifact
- (3) Feb–Mar 2020 outlier pair
- (4) Daily vs 5-min RV (noise, not sign)

Slope and R^2 Profiles: Paper (Red) vs. Ours (Blue)

Figure 3 Analog — Paper vs. Notebook (opposite slope signs)



Left: slope $b_1(h)$ — zero line cleanly separates both samples at all horizons. Right: adj. $R^2(h)$ — paper's hump peaks at $h=3$; ours is flat. Both shape and sign differ.

7

Robustness Checks

Do Alternative Specifications Change the Sign?

Robustness Checks: Three Alternative Specifications

Check 1: Excluding October 2008

- Slope at $h=1$: $-0.012 \rightarrow -0.008$ (less negative, but still negative)
- Slope at $h=3$: $-0.005 \rightarrow -0.001$ (near zero but still negative)
- Slope at $h=12$: $-0.003 \rightarrow -0.003$ (unchanged)
- Verdict: Oct 2008 partly drives $h=1$; Feb–Mar 2020 dominates $h=3+$. Sign reversal is structural.

Check 2: Volatility Form $\sqrt{IV} - \sqrt{RV}$

- Paper (quarterly): $t = 2.75$, $R^2 = 6.51\%$ (vs 2.86 / 6.82% in variance form)
- Our $h=1$: $t = -0.46$ | $h=3$: $t = -0.63$ | $h=6$: $t = -1.03$ | $h=12$: $t = -1.62$
- Square root compresses outlier magnitudes but does not change sign

Check 3: HAR-RV Expected VRP (EVRP) — Nearly Replicates the Paper!

- Replace raw RV_t with a HAR-RV forecast: $EVRP_t = IV_t - E_t[RV_{t+1}]$ (Corsi 2004 model, $R^2 = 0.35$)
- EVRP mean = 9.28, std = 25.50, $AR(1) = 0.23$ — smoothing restores partial signal persistence
- $h = 3$: $t = +2.79$, $R^2 = 5.91\%$ — nearly matches paper ($t = 2.86$, $R^2 = 6.82\%$)
- $h = 6$: $t = +2.99$, $R^2 = 6.38\%$ — even stronger. $h = 12$: $t = +2.47$, $R^2 = 4.97\%$

Robustness Conclusion

- EVRP (smoothed RV) recovers the sign. Raw VRP reversal = RV crash noise, not a structural IV-return change.

8

Honest Assessment

What This Replication Can and Cannot Support

Honest Assessment: Paper vs. Notebook

Result	Paper (1990–2007)	Our Notebook (2008–2026)	Verdict
VRP Sign + Persistence	Mean=18.3, AR(1)=0.49	Mean=9.0, AR(1)=-0.10	Sign ✓, persistence ✗
Quarterly prediction	$b_1=+0.47$, $t=+2.86$, $R^2=6.82\%$	$b_1=-0.005$, $t=-0.96$, $R^2=0.42\%$	SIGN REVERSED
Hump-shaped R^2	Peaks at $h=3$ then decays	Flat 0.4%–3.0%, no hump	Shape fails ✗
Backtest Sharpe	Implied $\sim 2\times$ uplift	$0.68 \rightarrow 0.75 (+0.07)$	Modest improvement $\sim$
Max Drawdown	Not reported	$-29.4\% \rightarrow -22.6\%$	Real improvement ✓
EVRP check	$t=2.27$, $R^2=4.27\%$ ($h=3$)	$t=2.79$, $R^2=5.91\%$ ($h=3$)	Replicates! ✓

What the Data Supports

- VRP is positive 85% of months (variance insurance premium is real). Avoiding VRP-negative months reduces max drawdown by 6.8 pp. HAR-RV smoothing recovers the paper's quarterly predictability signal (EVRP $t=2.79$).

Why Replication Fails: Structural Reasons and the Broader Lesson

Source of Discrepancy	Direction of Effect	Magnitude
Sample epoch (2008–2026 vs. 1990–2007)	Reverses sign (dominant)	Dominant
Oct 2008 partial-month artifact (4 trading days)	Worsens $h=1$ negative slope	Large
Feb–Mar 2020 outlier pair (COVID crash + recovery)	Drives negative $h=3$ to $h=12$	Large
Daily vs 5-minute realized variance	Weakens signal (noise), not sign	Moderate
Newey–West vs. Hodrick (1992) standard errors	Slightly different t-stats	Small
Fed Funds vs. 3-month T-bill as risk-free rate	Negligible in low-rate era	Very small

The Broader Lesson: Sample Sensitivity in Empirical Asset Pricing

- The paper's 1990–2007 regime had: (1) moderate persistent VRP ($AR1=0.49$), (2) no single catastrophic outlier pair, (3) fear spikes followed by recovery returns.
- Our 2008–2026 regime has: near-zero VRP persistence, extreme outlier pairs dominating the regression, and fear spikes that immediately precede crashes rather than predict recoveries.

Key Takeaways

- 1 VRP = IV – RV is theoretically motivated: Epstein–Zin preferences with stochastic vol-of-vol (q_t) create an equity premium component measured exactly by the VRP spread.
- 2 The paper finds peak predictability at $h=3$: $b_1=+0.47$, $t=+2.86$, $R^2=6.82\%$ for 1990–2007 — the VRP-return relationship is positive and significant at the quarterly horizon.
- 3 Our 2008–2026 replication finds the exact opposite: all slopes negative. Two outlier pairs (Oct–Nov 2008; Feb–Mar 2020) run the regression in reverse and dominate the 209-month sample.
- 4 VRP AR(1) collapses from 0.49 (paper) to -0.10 (ours) — near-zero persistence is inconsistent with the model's persistent q_t assumption and makes the signal unreliable.
- 5 The VRP Binary strategy improves Sharpe from 0.68 to 0.75 and cuts max drawdown from -29.4% to -22.6% , but by avoiding the 14.8% of crash months, not by predicting returns.
- 6 HAR-RV Expected VRP recovers the paper's finding: EVRP at $h=3$ gives $t=+2.79$, $R^2=5.91\%$ — smoothing realized variance restores predictability even in the post-2008 sample.
- 7 Regime sensitivity is the central lesson. Out-of-sample testing in a structurally different epoch is as informative as the original result — both findings are true within their own epoch.

Discussion Questions

1

The paper stopped at December 2007 to avoid the GFC. Is this a strength (clean signal, pre-crisis) or a limitation (implicit sample selection / p-hacking risk)? How would you test for this?

2

If you were managing a volatility fund in 2008, would you continue using the VRP signal after the October 2008 partial-month artifact? Design a real-time monitoring rule that would have flagged this observation.

3

The EVRP (using HAR-RV forecast of RV) nearly replicates the paper's result even in our post-2008 sample ($t=2.79$ vs. 2.86). Why might replacing realized with expected RV recover predictability?

4

VRP AR(1) drops from 0.49 (paper) to -0.10 (ours). Which structural changes — options market microstructure, QE/central bank intervention, index products growth — could explain this collapse in persistence?

5

Design a robustness test to determine whether the 2008 and 2020 outlier pairs are the sole drivers of the negative slope, or whether the post-2008 regime shift produces a different result even when those specific months are excluded.

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